

Project 2 : Flame Module

Introduction:

Pluto X2 has a valuable number of add-ons to achieve tasks that would be otherwise not be possible. One of these add-ons is a flame module. In this project we are going to interface it with Pluto X2.

Challenge:

Build a program to add fire sensing ability to Pluto X2. It should sense fire and take off accordingly.

Add-On used:

A flame sensor module is basically just a IR receiver module, which detects the fire by using the IR waves emitting from it.



Connection:

Connect the flame_module on the RC port on Primus V5/X2 board

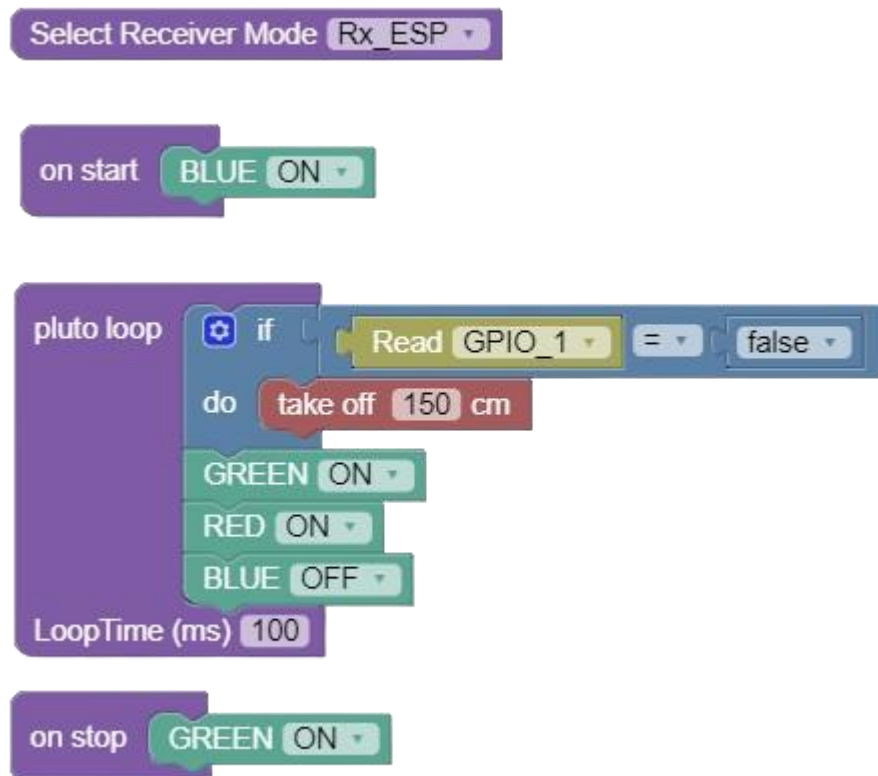
Primus V5/X2 pin	Peripheral
VCC on RC port	VCC on flame_module

GND on RC port	GND on flame_module
Signal on RC port	signal on flame_module

Approaching the Problem:

We utilise the input signal from flame sensor to evaluate the drone's take-off decision.

Solution:



Explanation of Solution:

On Start Loop:

We set Blue LED on.



Pluto Loop:

When we enter developer mode, pluto loop runs constantly and as an indicator of that, We make the indicator LED to Yellow. Flame sensor provides digital output. That is in form of 1's and 0's in other words HIGH/LOW at Pluto's input terminal. Under normal circumstances, sensor gives HIGH signal, but when it detects any flame/fire in its surrounding the Signal output of sensor goes LOW. This fact is utilised to make the decision of drone takeoff. Additionally the sensitivity of flame sensor can be changed by adjusting the potentiometer present on it.



On stop Loop: Green LED turns On





Fig: Flame sensor with Pluto

Try further!

- Utilise multiple flame sensor to locate fires .
- Add a hooter to make fire indicator drone.